MAYER-VIETORIS LONG EXACT SEQUENCE FOR A COVER BY TWO OPEN SUBSETS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed. The author's closing sentence omitting functoriality is reproduced as written, as is the bracketed name Mayer-Vietoris in the environment header.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot: sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The same-chapter pointer is located at its \label line inside the pinned chapter file; the cross-chapter pointer is located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim line carrying its canonical tag.

Allowed standard prerequisites (editorial).

- Restriction to an open of an injective module is surjective on sections. Same chapter, Lemma labelled lemma-injective-restriction-surjective (source line 715, Stacks tag 01EA); invoked at proof line 786. Supplies surjectivity of (1, -1) on sections of an injective module over U and V onto the intersections, which turns the degreewise sequence of an injective resolution into a short exact sequence of complexes.
- The long exact cohomology sequence of a short exact sequence of cochain complexes. Homology, Lemma labelled homology-lemma-long-exact-sequence-cochain (canonical Stacks tag 0117); invoked at proof line 799. Supplies the connecting maps and the long exact sequence once the short exact sequence of complexes is in hand.

Source reference. Original lemma 01EA: sources/stacks-cohomology/pinned-original-source.tex, source lines 715–715. Same-chapter pointer invoked at proof line 786 for surjectivity of (1, -1) on sections of an injective module; the assembly of the short exact sequence of complexes and the conclusion are retained in the counted span.

Source reference. Original lemma 0117: sources/stacks-chow/tags.txt, source lines 1353–1353. Cross-chapter standard prerequisite invoked once at proof line 799 to take the long exact cohomology sequence; the short exact sequence of complexes it is applied to is retained in the counted span.

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1 (Mayer-Vietoris)**.** *Let X be a ringed space. Suppose that $X = U \cup V$ is a union of two open subsets. For every $\mathcal{O}_X$-module $\mathcal{F}$ there exists a long exact cohomology sequence*

$$0 \to H^0(X, \mathcal{F}) \to H^0(U, \mathcal{F}) \oplus H^0(V, \mathcal{F}) \to H^0(U \cap V, \mathcal{F}) \to H^1(X, \mathcal{F}) \to \ldots$$

This long exact sequence is functorial in $\mathcal{F}$.

Proof. The sheaf condition says that the kernel of $(1, -1) : \mathcal{F}(U) \oplus \mathcal{F}(V) \to \mathcal{F}(U \cap V)$ is equal to the image of $\mathcal{F}(X)$ by the first map for any abelian sheaf $\mathcal{F}$. Lemma 01EA above implies that the map $(1, -1) : \mathcal{I}(U) \oplus \mathcal{I}(V) \to \mathcal{I}(U \cap V)$ is surjective whenever $\mathcal{I}$ is an injective $\mathcal{O}_X$-module. Hence if $\mathcal{F} \to \mathcal{I}^\bullet$ is an injective resolution of $\mathcal{F}$, then we get a short exact sequence of complexes

$$0 \to \mathcal{I}^\bullet(X) \to \mathcal{I}^\bullet(U) \oplus \mathcal{I}^\bullet(V) \to \mathcal{I}^\bullet(U \cap V) \to 0.$$

Taking cohomology gives the result (use Homology, Lemma 0117). We omit the proof of the functoriality of the sequence. □